Product Information **Ultramid®**

A3X2G10

04/2025 **PA66-GF50 FR(52)**

Product Information

Glass fibre fibre reinforced injection moulding grade with improved flame retardance. Flame retardant based on red phosphorus; giving outstanding electrical properties and very high stiffness and strength.
The products can also be offered as BMBcert™ and/or Ccycled™. Due to the Massbalance approach the product properties do not change.

Physical form and storage

The product is supplied in the form of granules with a bulk density of approx. 0.7 g/cm³. Standard packs are bag and bulk container (octagonal IBC=intermediate bulk container made from corrugated board with a liner bag). Other packaging materials and shipping in road or rail silo wagons are possible by agreement. The containers should only be opened immediately before processing or drying. To ensure that the delivered product absorbs as little moisture as possible, the containers should be stored in dry rooms and always carefully closed again after partial quantities have been withdrawn. In principle, the product can be stored for a long period of time. Containers stored in cold rooms should be equalized to ambient temperature before opening in order to avoid condensation on the granules. Regardless of the storage conditions, the product should be pre-dried according to our recommendations and the machine should preferably be loaded using a closed conveyor system.

Product safety

In case processing is done under conditions as recommended (cf. processing data sheet) melts are thermally stable and do not generate hazards by molecular degradation or the evolution of gases and vapors. Like all thermoplastic polymers the product decomposes on exposure to excessive thermal load, e.g. when it is overheated or as a result of cleaning by burning off. Further information is available from the safety data sheet.

Safety instructions

Provide suitable exhaust ventilation at the drying process and in the area surrounding the melt outlet of processing machines.
Closed containers should only be opened in well-ventilated areas.
Ensure thorough ventilation of stores and work areas.

When incorrectly processing an unpleasant odour can be produced, especially when the recommended processing parameters are exceeded.
Check
- Moisture content of pellets
- Melt temperature
- Residence time

When there is a strong odour, immediately check processing parameters, ventilate the area well and recheck moisture content of material. If necessary stop processing and redry the material.

Any short stoppages in production, it is recommended that you inject material into the mould not purge an air shot.
Any molten material drooling from the machine nozzle or hot runner nozzles can self-ignite when in open atmosphere.
It is therefore advisable to dispose of purgings etc into water containers.

Note

The data contained in this publication are based on our current knowledge and experience. In view of the many factors that may affect processing and application of our product, these data do not relieve processors from carrying out their own investigations and tests; neither do these data imply any guarantee of certain properties, nor the suitability of the product for a specific purpose. Any descriptions, drawings, photographs, data, proportions, weights etc. given herein may change without prior information and do not constitute the agreed contractual quality of the product. It is the responsibility of the recipient of our products to ensure that any proprietary rights and existing laws and legislation are observed.
In order to check the availability of products please contact us or our sales agency.

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67056 Ludwigshafen, Germany

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Product Information

We create chemistry

Typical values for uncoloured product at 23 °C[1]	Test method	Unit	Values[2]
Properties			
Polymer abbreviation	-	-	**PA66-GF50 FR(52)**
Density	ISO 1183	kg/m³	**1600**
Viscosity number (0.5% in 96% H_2SO_4)	ISO 307, 1157, 1628	cm³/g	**140**
Water absorption, saturation in water at 23°C	similar to ISO 62	%	**3.7 - 4.3**
Moisture absorption, equilibrium 23°C/50% r.h.	similar to ISO 62	%	**0.7 - 1.1**
Processing			
Melting temperature, DSC	ISO 11357-1/-3	°C	**260**
MVR 275 °C/5 kg	ISO 1133	cm³/10min	**25**
Melt temperature, injection moulding/extrusion	-	°C	**290 - 300**
Mould temperature, injection moulding	-	°C	**80 - 90**
Moulding shrinkage, constrained [3]	-	%	**0.4**
Flammability (UL yellow card see attachment)			
Glow Wire Flammability Index, GWFI at d = 1 mm thickness	IEC 60695-2-12	°C	**960**
Oxygen index	ISO 4589-1/-2	%	**27**
Specific optical smoke density	EN ISO 5659-2: 2007	-	**184**
Toxicity of smoke CIT NLP acc. to CEN/TS 45545-2	NF X70-100-1/-2	-	**0.36**
Mechanical properties			**dry / cond.**
Tensile modulus	ISO 527-1/-2	MPa	**16000 / 12000**
Stress at break	ISO 527-1/-2	MPa	**180 / 130**
Strain at break	ISO 527-1/-2	%	**2 / 3**
Tensile creep modulus, 1000 h, strain ≤ 0.5%, 23°C	ISO 899-1	MPa	*** / 5400**
Flexural modulus	ISO 178	MPa	**13000 / -**
Charpy unnotched impact strength (23°C)	ISO 179/1eU	kJ/m²	**55 / 55**
Charpy unnotched impact strength (-30°C)	ISO 179/1eU	kJ/m²	**50 / -**
Charpy notched impact strength (23°C)	ISO 179/1eA	kJ/m²	**13 / 16**
Charpy notched impact strength (-30°C)	ISO 179/1eA	kJ/m²	**11 / -**
Izod notched impact strength (23°C)	ISO 180/A	kJ/m²	**14 / 20**
Thermal properties			
Deflection temp. under load 1.8 MPa (HDT A)	ISO 75-1/-2	°C	**250**
Deflection temp. under load 0.45 MPa (HDT B)	ISO 75-1/-2	°C	**250**
Max. service temperature (short cycle operation)	-	°C	**220**
Temperature index at 50% loss of tensile strength after 5000 h	IEC 60216	°C	**145**
Temperature index at 50% loss of tensile strength after 20000 h	IEC 60216	°C	**125**
Coefficient of linear thermal expansion, longitudinal (23-80)°C	ISO 11359-1/-2	E-6/K	**17**
Coefficient of linear thermal expansion, transverse (23-80)°C	ISO 11359-1/-2	E-6/K	**66**
Thermal conductivity	DIN 52612-1	W/(m K)	**0.35**
Specific heat capacity	-	J/(kg*K)	**1300**
Electrical properties			**dry / cond.**
Relative permittivity (1 MHz)	IEC 62631-2-1	-	**3.6 / 5**
Dissipation factor (1 MHz)	IEC 62631-2-1	E-4	**200 / -**
Volume resistivity	IEC 62631-3-1	Ohm*m	**1E13 / 1E10**
Surface resistivity	IEC 62631-3-2	Ohm	**- / 1E13**
Comparative tracking index, CTI, test liquid A	IEC 60112	-	**600**
Electric strength K20/K20, (60*60*1 mm³)	IEC 60243-1	kV/mm	**33 / 30**

Footnotes
1) If product name or properties don't state otherwise.
2) The asterisk symbol '*' signifies inapplicable properties.
3) Test box with central gating, dimensions of base (107*47*1,5) mm, processing condition: TM = 320°C (unreinforced) or 330°C (reinforced), TW = 80°C

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Ultramid® A3X2G10

UL - Yellow Card

Component - Plastics E41871

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Performance Materials Europe, PMD/EX - H201, Ludwigshafen 67056 DE

A3X2G10(f2), A3X3G10(f2), A3X2G10 LF(f2), A3X2G10 BMB(f2)

Polyamide 66 (PA66) "Ultramid", furnished as pellets

Color	Min. Thk (mm)	Flame Class	HWI	HAI	RTI Elec (°C)	RTI Imp (°C)	RTI Str (°C)
NC, BK	0.40	V-2	4	1	65	65	65
	0.60	V-2	1	1	105	105	120
NC, BK, GY	0.75	V-2	1	1	110	115	130
NC, GY	1.5	V-0	0	1	115	115	130
	3.0	V-0	0	1	115	115	130
BK	1.5	V-0, 5VA	0	1	115	115	130
	3.0	V-0, 5VA	0	1	115	115	130

Comparative Tracking Index (CTI): 0

Inclined Plane Tracking (IPT) kV: -

Dielectric Strength (kV/mm): 21

Volume Resistivity (10^xohm-cm): 13

High-Voltage Arc Tracking Rate (HVTR): 0

Surface Resistivity (10^xohms/square): -

Dimensional Change (%): 0

High Volt, Low Current Arc Resis (D495): 5

(f2) - Subjected to one or more of the following tests: Ultraviolet Light, Water Exposure or Immersion in accordance with UL 746C, where the acceptability for outdoor use is to be determined by UL.

ANSI/UL 94 small-scale test data does not pertain to building materials, furnishings and related contents. ANSI/UL 94 small-scale test data is intended solely for determining the flammability of plastic materials used in the components and parts of end-product devices and appliances, where the acceptability of the combination is determined by UL.

Report Date: 1974-10-24

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IEC and ISO Test Methods

Test Name	Test Method	Units	Thk (mm)	Value
Flammability	IEC 60695-11-10, IEC 60695-11-20	Class (color)	0.40	V-2 (NC, BK)
			0.60	V-2 (NC, BK)
			0.75	V-2 (NC, BK, GY)
			1.5	V-0 (NC, GY)
			3.0	V-0 (NC, GY)
			1.5	V-0, 5VA (BK)
			3.0	V-0, 5VA (BK)
Glow-Wire Flammability (GWFI)	IEC 60695-2-12	°C	-	-
Glow-Wire Ignition (GWIT)	IEC 60695-2-13	°C	-	-
IEC Comparative Tracking Index	IEC 60112	Volts (Max)	-	-
IEC AC Dielectric Strength (AC DS)	IEC 60243-1	kV/mm	-	-
IEC DC Dielectric Strength (DC DS)	IEC 60243-2	kV/mm	-	-

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UL - Yellow Card

IEC Volume Resistivity (VR)	IEC 62631-3-1	10x ohm-m	-	-
IEC Surface Resistivity (SR)	IEC 62631-3-2	10x ohms	-	-
IEC Inclined Plane Tracking (IPT)	IEC 60587	kV	-	-
IEC Ball Pressure	IEC 60695-10-2	°C	-	-
ISO Heat Deflection (1.80 MPa)	ISO 75-2	°C	-	-
ISO Tensile Strength	ISO 527-2	MPa	-	-
ISO Flexural Strength	ISO 178	MPa	-	-
ISO Tensile Impact	ISO 8256	kJ/m2	-	-
ISO Izod Impact	ISO 180	kJ/m2	-	-
ISO Charpy Impact	ISO 179-1	kJ/m2	-	-

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A3X2G10(f1), A3X3G10(f1), A3X2G10 LF(f1), A3X2G10 BMB(f1)

Polyamide 66 (PA66) "Ultramid", furnished as pellets

Color	Min. Thk (mm)	Flame Class	HWI	HAI	RTI Elec (°C)	RTI Imp (°C)	RTI Str (°C)
BK	1.5	V-0, 5VA	0	1	115	115	130
	3.0	V-0, 5VA	0	1	115	115	130

Comparative Tracking Index (CTI):	0	Inclined Plane Tracking (IPT) kV:	-
Dielectric Strength (kV/mm):	21	Volume Resistivity (10^xohm-cm):	13
High-Voltage Arc Tracking Rate (HVTR):	0	Surface Resistivity (10^xohms/square):	-
Dimensional Change (%):	0	High Volt, Low Current Arc Resis (D495):	5

(f1) - Suitable for outdoor use with respect to exposure to Ultraviolet Light, Water Exposure and Immersion in accordance with UL 746C.

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IEC and ISO Test Methods

Test Name	Test Method	Units	Thk (mm)	Value
Flammability	IEC 60695-11-10, IEC 60695-11-20	Class (color)	1.5	V-0, 5VA (BK)
			3.0	V-0, 5VA (BK)
Glow-Wire Flammability (GWFI)	IEC 60695-2-12	°C	-	-
Glow-Wire Ignition (GWIT)	IEC 60695-2-13	°C	-	-
IEC Comparative Tracking Index	IEC 60112	Volts (Max)	-	-
IEC AC Dielectric Strength (AC DS)	IEC 60243-1	kV/mm	-	-
IEC DC Dielectric Strength (DC DS)	IEC 60243-2	kV/mm	-	-
IEC Volume Resistivity (VR)	IEC 62631-3-1	10x ohm-m	-	-
IEC Surface Resistivity (SR)	IEC 62631-3-2	10x ohms	-	-
IEC Inclined Plane Tracking (IPT)	IEC 60587	kV	-	-
IEC Ball Pressure	IEC 60695-10-2	°C	-	-
ISO Heat Deflection (1.80 MPa)	ISO 75-2	°C	-	-
ISO Tensile Strength	ISO 527-2	MPa	-	-
ISO Flexural Strength	ISO 178	MPa	-	-
ISO Tensile Impact	ISO 8256	kJ/m2	-	-

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ISO Izod Impact	ISO 180	kJ/m2	-	-
ISO Charpy Impact	ISO 179-1	kJ/m2	-	-

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